

Introduction to SQL

We have already learnt that databases and DBMS are key to organising and analysing data for business uses.

From here on, let's get busy working around with databases using SQL!

- SQL stands for Structured Query Language
- SQL is used to perform operations on Relational DBMS.
- SQL is declarative. Hence, easy to learn.

SQL provides multiple clauses (commands) to perform various operations like create, retrieve, update and delete the data.

The first step towards working with the database would be creating a table.

Data Types

Following data types are frequently used in SQL.

Data Type	Syntax
Integer	INTEGER / INT
Float	FLOAT
String	VARCHAR
Text	TEXT
Date	DATE
Time	TIME
Datetime	DATETIME
Boolean	BOOLEAN

Note

1. Boolean values are stored as integers 0 (FALSE) and 1 (TRUE).
2. Date object is represented as: 'YYYY-MM-DD'
3. Datetime object is represented as: 'YYYY-MM-DD HH:MM:SS'

Create Table

Creates a new table in the database.

Syntax

SQL

```
CREATE TABLE table_name (  
    column1 type1,  
    column2 type2,  
    ...  
);
```

Here,
type1 and type2 in the syntax are the datatypes
of column1 and column2 respectively.

Example

Create a

playertable to store the following details of players.

column_name	data_type
name	VARCHAR(200)
age	INT/INTEGER
score	INT/INTEGER

SQL

```
CREATE TABLE player (  
    name VARCHAR(200),  
    age INTEGER,  
    score INTEGER  
);
```

We can check the details of the created table at any point in time using the PRAGMAcommand.

Try it Yourself!

Assume that we have to build a database that stores all the information about the students in a school, subjects, exam schedules, etc. Lets build a few tables to store the data!

1. Create astudenttable to store the following details of students.

details	data_type
name	VARCHAR(200)
date_of_birth	DATE
address	TEXT

Check if the tables are created in the database after you run the query

1. Create anexam_scheduledtable to store the information about exams.

details	data_type
name	VARCHAR(200)
course	VARCHAR(200)
exam_date_time	DATETIME
duration_in_sec	INT
pass_percentage	FLOAT

Check if the tables are created in the database after you run the query

PRAGMA

PRAGMA TABLE_INFOcommand returns the information about a specific table in a database.

Syntax

SQL

```
PRAGMA TABLE_INFO(table_name);
```

Example

Let's find out the information of the employeetable that's present in the database.

SQL

```
PRAGMA TABLE_INFO(employee);
```

Output

cid	name	type	notnull	dflt_value	pk
0	employee_id	INTEGER	0		0
1	name	VARCHAR (200)	0		0
2	salary	INTEGER	0		

Note:

dflt_value is the default value of the column. If a column does not have a default value specified by the user while creating, the dflt_value column will be NULL.

Note

If the given table name does not exist,PRAGMA TABLE_INFOdoesn't give any result.

Try it Yourself!

Try checking out the information of the tables that you have created above.